Unit 5 Exercises

4SC000 Optimal Control and Reinforcement Learning

Duarte Antunes

A runnable Python companion to these exercises – Riccati recursion (both finite- and infinite-horizon) plus the double-integrator and DC-motor examples – is available in `code/unit5/` (see its `README.md`).

Core exercises

Problem 5.1 Consider a first order discrete-time linear system

$$x_{k+1} = \alpha x_k + u_k, \quad k \in \mathbb{N}_0,$$

and the cost

$$\sum_{k=0}^{q} x_k^2 + \gamma u_k^2.$$

(a) Considering $\alpha = 1$, $\gamma = 1$ and $q = 2$, determine the policy for u_0, u_1, and u_2 that minimizes the cost.
(b) Considering $\alpha = 1$, $\gamma = 1$ and $q = \infty$, determine the policy for $\{u_k, k \in \mathbb{N}_0\}$ that minimizes the cost.
(c) Suppose now that $|\alpha| < 1$ and $q = \infty$. Provide and interpret the policy that minimizes the cost when $\gamma \to \infty$.

Solution

(a) For $\alpha = 1$, $\gamma = 1$, $q = 2$, we have $x_{k+1} = x_k + u_k$ and $\sum_{k=0}^{2} \underbrace{x_k^2 + u_k^2}_{g(x_k,u_k)} + \underbrace{0}_{J_3(x_3)}$, a standard stage decision problem. Applying DP:
Step 1. $J_2(x_2) = \min_{u_2} x_2^2 + u_2^2 = x_2^2$, where $u_2 = \mu(x_2) = 0 \ \forall x_2$.
Step 2. $J_1(x_1) = \min_{u_1} x_1^2 + u_1^2 + (x_1 + u_1)^2 = \min_{u_1} 2x_1^2 + 2u_1^2 + 2x_1u_1$. Differentiating and equating to zero: $4u_1 + 2x_1 = 0$, so $u_1 = -\frac{1}{2}x_1$ and $J_1(x_1) = \frac{3}{2}x_1^2$.
Step 3. $J_0(x_0) = \min_{u_0} x_0^2 + u_0^2 + \frac{3}{2}(x_0 + u_0)^2 = \min_{u_0} \frac{5}{2}x_0^2 + \frac{5}{2}u_0^2 + 3x_0u_0$. Differentiating and equating to zero: $5u_0 + 3x_0 = 0$, so $u_0 = -\frac{3}{5}x_0$ and $J_0(x_0) = 1.6x_0^2$.
Alternatively, iterate the (scalar) Riccati equation $p_k = p_{k+1} + 1 - \frac{p_{k+1}^2}{1+p_{k+1}}$ for $k \in \{2, 1, 0\}$, $p_3 = 0$, with gains $K_k = -\frac{p_{k+1}}{1+p_{k+1}}$ (costs-to-go $x_k^2 p_k$). This gives $p_2 = 1$, $K_2 = 0$; $p_1 = \frac{3}{2}$, $K_1 = -\frac{1}{2}$; $p_0 = 1.6$, $K_0 = -\frac{3}{5}$ – matching the by-hand recursion.
(b) Since the model is controllable and the cost matrix is positive definite, the backward iteration converges to a limit p satisfying the algebraic Riccati equation $p = p + 1 - \frac{p^2}{1+p}$, i.e. $p^2 - p - 1 = 0$, with roots $p = \frac{1 \pm \sqrt{5}}{2}$. Taking the positive-definite root, $p = \frac{1+\sqrt{5}}{2} = 1.618$, so the optimal cost is $x_0^2 p$ and the optimal policy is $u_k = Kx_k$, $k \in \mathbb{N}_0$, where $K = -\frac{p}{1+p} = -\frac{1+\sqrt{5}}{3+\sqrt{5}} = -0.618$.
(c) $P(\gamma)$ satisfies $p = \alpha^2 p + 1 - p^2\alpha^2/(\gamma + p)$ and converges to $\frac{1}{1-\alpha^2}$ as $\gamma \to \infty$. Thus $K(\gamma) = -\frac{\alpha^2 P(\gamma)^2}{\gamma + P(\gamma)}$ converges to zero, so the optimal policy is $u_k = 0$ for every k. Interpretation: if control is expensive and the system is already stable ($\alpha < 1$), it is optimal to just let it run open loop. The resulting cost is

obtained from iterating $x_{k+1} = \alpha x_k$: $\sum_{k=0}^{\infty} x_k^2 = \sum_{k=0}^{\infty} (\alpha^2)^k x_0^2 = \frac{1}{1-\alpha^2} x_0^2$.

Problem 5.2 Consider the standard linear quadratic problem

$$\min \sum_{k=0}^{h-1} x_k^\top Q x_k + u_k^\top R u_k + 2 x_k^\top S u_k + g_h(x_h) \qquad x_{k+1} = A x_k + B u_k$$

for

$$Q = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad S = \begin{bmatrix} 0.1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 & 0 \\ -3 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 2 \end{bmatrix}$$

and $g_h(x_h) = x_h^\top Q_h x_h$, $Q_h = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

(i) Provide the gains K_k of the optimal policy $u_k = K_k x_k$, $k \in \{0, 1, 2, 3, 4\}$, when $h = 5$, by iterating the Riccati equations.
(ii) Using the same recursion, obtain the optimal policy $u_k = K x_k$ when $g_h(x_h) = 0$ and $h = \infty$.
(iii) Confirm the results of (ii) using the algebraic Riccati equation directly (e.g. via `dlqr.m` in Matlab, or `scipy.linalg.solve_discrete_are` in Python).

i Solution

(i) Iterating the Riccati/gain equations backward from $P_5 = Q_5 = Q_h$ gives

$$K_0 = \begin{bmatrix} 2.8560 & -3.3810 & -0.9790 \\ 0.0263 & 0.0605 & -1.4179 \end{bmatrix} \quad K_1 = \begin{bmatrix} 2.8494 & -3.3649 & -0.9777 \\ 0.0271 & 0.0589 & -1.4180 \end{bmatrix} \quad K_2 = \begin{bmatrix} 2.8191 & -3.2769 & -0.9749 \\ 0.0344 & 0.0450 & -1.4161 \end{bmatrix}$$

$$K_3 = \begin{bmatrix} 2.7408 & -3.0849 & -0.9967 \\ 0.0829 & -0.0448 & -1.3211 \end{bmatrix} \quad K_4 = \begin{bmatrix} 1.9667 & -2.6667 & -0.6667 \\ 0 & 0 & 0 \end{bmatrix}$$

The optimal policy is $u_k = K_k x_k$, $k \in \{0, 1, 2, 3, 4\}$.
(ii) Iterating with a large horizon (e.g. $h = 20$) the gains converge, $\bar{K} = K_0 = \begin{bmatrix} 2.8570 & -3.3830 & -0.9792 \\ 0.0262 & 0.0607 & -1.4179 \end{bmatrix}$. The optimal policy is $u_k = \bar{K} x_k$.
(iii) Solving the algebraic Riccati equation directly reproduces the same gains as in (ii).

Additional exercises

Problem 5.3 A schematic model of a DC motor is depicted below. The torque generated by the motor is assumed proportional to the armature current, $T = K_t i$, and the back emf e is assumed proportional to the angular velocity of the shaft, $e = K_e \dot{\theta}$. Newton's law and Kirchhoff's voltage law give

$$J\ddot{\theta} + b\dot{\theta} = K_t i \qquad L\frac{di}{dt} + Ri = V - K_e \dot{\theta}$$

Assuming the inductance is very small, $L \approx 0$, this reduces to $J\ddot{\theta} + (b + \frac{K_t K_e}{R})\dot{\theta} = \frac{K_t}{R} V$. Parameters: $J = 0.01$, $b = 0.1$, $K_e = 0.01$, $K_t = 0.01$, $R = 1$, $L = 0$.

(i) Find a state-space representation for $x(t) := [\theta(t) \;\; \dot{\theta}(t)]^\top$.

(ii) Discretize this model for $\tau = 0.1$, giving $x_{k+1} = A_d x_k + B_d u_k$ with $x_k := x(\tau k)$.

(iii) Suppose we want to regulate the state to zero while minimizing $J = \sum_{k=0}^{\infty} x_k^\top Q x_k + u_k^\top R u_k$, $Q = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$, $R = 0.1$. (a) Compute the LQR gain K^* ($u_k = K^* x_k$) and the cost matrix P^* ($x_0^\top P^* x_0$). (b) Compute the control that regulates the state to zero in the minimum number of steps (deadbeat, cf. Problem 5.5), as a function of x_0; show the resulting cost is $x_0^\top P_f x_0$ for some P_f. (c) Compare the costs obtained in (a) and (b).

Solution

(i) $\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & -\alpha \end{bmatrix} x + \begin{bmatrix} 0 \\ \beta \end{bmatrix} V$, where $\alpha = \frac{b + K_t K_e / R}{J} = 10.01$ and $\beta = \frac{K_t}{RJ} = 1$.

(ii) With $A = \begin{bmatrix} 0 & 1 \\ 0 & -\alpha \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ \beta \end{bmatrix}$: $A_d = e^{A\tau} = \begin{bmatrix} 1.0000 & 0.0632 \\ 0 & 0.3675 \end{bmatrix}$, $B_d = \int_0^\tau e^{As} ds\, B = \begin{bmatrix} 0.0037 \\ 0.0632 \end{bmatrix}$.

(iii)(a) $K = -\begin{bmatrix} 3.9698 & 1.3993 \end{bmatrix}$, $P = \begin{bmatrix} 57.4811 & 4.6278 \\ 4.6278 & 6.0073 \end{bmatrix}$.

(iii)(b) From the controllability-matrix formula of Problem 5.5, $\begin{bmatrix} K_0 \\ K_1 \end{bmatrix} = \begin{bmatrix} A_d B_d & B_d \end{bmatrix}^{-1} (-A_d^2)$, which leads to $x_1 = (A_d + B_d K_0) x_0$ and $x_2 = 0$. The cost is $x_0^\top Q x_0 + x_0^\top K_0^\top R K_0 x_0 + x_0^\top (A_d + B_d K_0)^\top Q (A_d + B_d K_0) x_0 + x_0^\top K_1^\top R K_1 x_0 = x_0^\top P_f x_0$, where $P_f = Q + K_0^\top R K_0 + (A_d + B_d K_0)^\top Q (A_d + B_d K_0) + K_1^\top R K_1$. Numerically, $K_0 = \begin{bmatrix} -158.2638 & -12.4309 \end{bmatrix}$, $K_1 = \begin{bmatrix} 58.1638 & 2.4309 \end{bmatrix}$, $P_f = \begin{bmatrix} 3345.4 & 231.78 \\ 231.78 & 21.92 \end{bmatrix}$.

(iii)(c) $x_0^\top P^* x_0 \le x_0^\top P_f x_0$ for every x_0, since $P_f - P^*$ is positive definite – driving the state to zero as fast as possible (deadbeat) is never cheaper than the LQR-optimal trade-off.

Problem 5.4 In this problem we wish to prove that the cost $\sum_{k=0}^{\infty} x_k^\top Q x_k$, $Q = Q^\top$, for an asymptotically stable discrete-time system $x_{k+1} = A x_k$, is given by $x_0^\top P x_0$, where P solves the Lyapunov equation $A^\top P A - P = -Q$.

(a) Prove that $\sum_{k=0}^{T} a^k = \frac{1 - a^{T+1}}{1 - a}$ for $a \in (-1, 1)$, by multiplying the left-hand side by $\frac{1-a}{1-a}$.
(b) Use (a) to prove the desired result for $A \in \mathbb{R}$, $|A| < 1$.
(c) Generalize (a)-(b) to $A \in \mathbb{R}^{n \times n}$ with eigenvalues inside the unit circle.
(d) Confirm that if A and Q are replaced by $(A + BK)$ and $Q + K^\top R K$, for $K = -(B^\top P B + R)^{-1} B^\top P A$, the Lyapunov equation coincides with the Riccati equation $P = A^\top P A + Q - A^\top P B (B^\top P B + R)^{-1} B^\top P A$. Explain why.

Solution

(a) $\sum_{k=0}^{T} a^k = \frac{1-a}{1-a} \sum_{k=0}^{T} a^k = \frac{1}{1-a} \sum_{k=0}^{T} (a^k - a^{k+1}) = \frac{1}{1-a}(1 - a^{T+1})$ (telescoping sum).

(b) $J = \sum_{k=0}^{T} x_k^\top Q x_k = x_0^\top [\sum_{k=0}^{T} (A^k)^\top Q A^k] x_0$. For scalar A, this is $x_0^2 Q \sum_{k=0}^{T} (A^2)^k$. As $T \to \infty$, using (a) with $a = A^2$, $\sum_{k=0}^{\infty} (A^2)^k = \frac{1}{1 - A^2}$, so $J = x_0^2 Q \frac{1}{1 - A^2}$. The Lyapunov equation $A^\top P A - P = -Q$ gives $Q = P(1 - A^2)$, hence $J = x_0^2 P$.

(c) $J = x_0^\top [\sum_{k=0}^{T} (A^k)^\top Q A^k] x_0$; substituting $Q = P - A^\top P A$ (from the Lyapunov equation) telescopes the sum so that only the $k = 0$ and $k = T + 1$ terms survive: $J = x_0^\top [P - (A^{T+1})^\top P A^{T+1}] x_0$, and the last term vanishes as $T \to \infty$ since A is asymptotically stable.

(d) Direct substitution. If we apply the LQR policy $u_k = K x_k$ to $\min \sum_{k=0}^{\infty} x_k^\top Q x_k + u_k^\top R u_k$ for $x_{k+1} = A x_k + B u_k$, we get the cost $\sum_{k=0}^{\infty} x_k^\top \underbrace{(Q + K^\top R K)}_{\bar{Q}} x_k$ for the closed loop $x_{k+1} = \underbrace{(A + BK)}_{\bar{A}} x_k$ – i.e. the Riccati equation is exactly the Lyapunov equation for the closed-loop cost, confirming (via (a)-(c)) that $x_0^\top P x_0$ is indeed the LQR optimal cost.

Problem 5.5 Consider the discrete-time linear time-invariant system $x_{k+1} = A x_k + B u_k$, $x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}$,

$k \in \mathbb{N}_0$, with arbitrary initial condition x_0. Suppose we wish to drive the state to a desired final value $\bar{x}$. Show that the control values

$$\begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{n-1} \end{bmatrix} = \begin{bmatrix} A^{n-1}B & A^{n-2}B & \cdots & B \end{bmatrix}^{-1} (\bar{x} - A^n x_0)$$

result in $x_n = \bar{x}$, provided the matrix on the right-hand side is invertible (in which case the system is said to be controllable).

i Solution

From $x_{k+1} = Ax_k + Bu_k$ we conclude that $x_n = A^n x_0 + \begin{bmatrix} A^{n-1}B & A^{n-2}B & \cdots & B \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{n-1} \end{bmatrix}$. Setting $x_n = \bar{x}$ and solving for the control vector (assuming invertibility) gives exactly the stated formula.

Problem 5.6 The goal of this exercise is to provide examples of linear quadratic regulation problems $\sum_{k=0}^{\infty} x_k^\top Q x_k + u_k^\top R u_k$, $x_{k+1} = Ax_k + Bu_k$, for which the optimal control policy cannot be obtained by solving the algebraic Riccati equation $P = A^\top PA + Q - (A^\top PB)(B^\top PB + R)^{-1}(B^\top PA)$.

(i) Provide an example of a pair (A, B) for which there does not exist a positive definite solution to the algebraic Riccati equation for any Q and R.
(ii) Provide an example of a pair (A, Q) for which there does not exist a positive definite solution for any B and R.

i Solution

(i) Take $B = 0$ and A not Hurwitz (e.g. $A = 2$, a scalar). The model reduces to $x_{k+1} = Ax_k$, whose trajectories blow up for any nonzero initial condition ((A, B) is not controllable/stabilizable). The algebraic Riccati equation becomes $P = 4P + Q$, whose unique solution $P = -Q/3$ is not positive definite for any positive semi-definite Q.
(ii) Take $Q = 0$, $A = 0$: the algebraic Riccati equation becomes $P = 0$, whose solution is not positive definite (the pair (A, N) with $Q = NN^\top$ fails observability/detectability).

Problem 5.7 The goal of this exercise is to show that we do not need R to be positive definite to obtain the optimal policy for the standard linear regulator problem with the algebraic Riccati equation. Consider the setting of Problem 5.2 with

$$A = \begin{bmatrix} 1/2 & 1 \\ 1 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad R = 0 \qquad S = 0 \qquad Q = I \qquad Q_h = I$$

(i) Provide the gains K_k of the optimal policy $u_k = K_k x_k$, $k \in \{0, 1, 2, 3, 4\}$, when $h = 5$, by iterating the Riccati equations.
(ii) Using the same recursion, obtain the optimal policy $u_k = Kx_k$ for $h = \infty$.
(iii) Try to solve the algebraic Riccati equation directly with a solver that requires $R > 0$ (e.g. `dlqr.m`) and see that it does not work.

Solution

(i) Iterating the Riccati recursion with these parameters gives

$$K_0 = [-1.1328 \quad -3.2656] \quad K_1 = [-1.1328 \quad -3.2655] \quad K_2 = [-1.1324 \quad -3.2647] \quad K_3 = [-1.1250 \quad -3.2500] \quad K_4 =$$

The optimal policy is $u_k = K_k x_k$, $k \in \{0, 1, 2, 3, 4\}$.
(ii) For a large horizon (e.g. $h = 20$) the gains converge to $\bar{K} = K_0 = [-1.1328 \quad -3.2656]$. The optimal policy is $u_k = \bar{K} x_k$.
(iii) A solver requiring R positive definite (like `dlqr.m`) fails here, but (ii) shows this is not actually needed for the algebraic Riccati equation itself to admit a solution and define the optimal policy – the requirement $R > 0$ is a sufficient condition used by some solvers/proofs, not a necessary one.